

Assignment III

AI Compute Platforms, Applications and Trends

Task 1: Create the Python Application

Create a Python program that:

1. Accepts two numbers from the user.
2. Calculates and displays their addition result.
3. Test the program locally to confirm that it works correctly.

Task 2: Create Docker Image Version 1

1. Create a Dockerfile for the Python application.
2. Build a Docker image with the following name and tag:
`calc-app:v1`
3. Verify that the image was created successfully.
4. Create and run a container using the `calc-app:v1` image.
5. Test the application inside the container.

Since the program requires user input, run the container in interactive mode.

Task 3: Push Version 1 to Docker Hub

1. Create a new public repository named `calc-app` on Docker Hub.
2. Tag the local image using your Docker Hub username:
`<dockerhub-username>/calc-app:v1`
3. Log in to Docker Hub from the terminal.
4. Push the `v1` image to the Docker Hub repository.
5. Verify that the image appears in your Docker Hub repository.

Task 4: Modify the Application

Modify the existing Python program so that it:

1. Accepts two numbers from the user.
2. Calculates and displays the addition result.
3. Calculates and displays the multiplication result.
4. Test the modified program locally.

Task 5: Create Docker Image Version 2

1. Build a new Docker image from the modified application.
2. Name and tag the new image as:
`calc-app:v2`
3. Create and run a new container using the `calc-app:v2` image.
4. Test both addition and multiplication operations.

Task 6: Push Version 2 to Docker Hub

If the modified application runs successfully:

1. Tag the new image as:

`<dockerhub-username>/calc-app:v2`

2. Push it to the **same Docker Hub repository** used for Version 1.
3. Verify that the repository contains both tags:

`v1`

`v2`

Expected Results

- `v1` displays the addition result.
- `v2` displays both addition and multiplication results.
- Both images are available in the same Docker Hub repository with different tags.
- The containers must be run and tested successfully.

Submission Include:

Student Name:

Student ID:

Assignment Number:

Docker hub link :

The PDF must include the commands used and screenshots of the successful results. Submit the following evidence:

1. Version 1 – Local Test and Docker Test

Screenshot showing the Python program running locally with two input numbers and the addition result.

Screenshot showing docker images with calc-app:v1.

- Screenshot showing calc-app:v1 running successfully in interactive mode and displaying the addition result.

2. Version 1 – Docker Hub

Screenshot of the Docker Hub calc-app repository showing the v1 tag.

3. Version 2 – Docker Test

Screenshot showing docker images with both calc-app:v1 and calc-app:v2.

Screenshot showing calc-app:v2 running successfully and displaying both addition and multiplication results.

4. Final Docker Hub Repository

Screenshot of the same calc-app Docker Hub repository showing both v1 and v2 tags .

5. Source Files

Include the final main.py and Dockerfile contents in the report.

Important: Screenshots must clearly show the student's terminal commands and their corresponding output. Do not submit screenshots containing only commands without execution results.